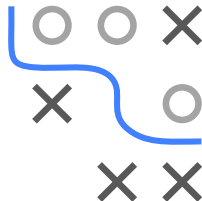


Introduction to Machine Learning

Evaluation ROC Basics



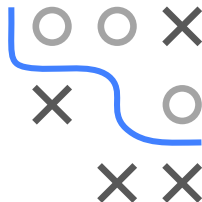
Learning goals

- Understand why accuracy is not an optimal performance measure for imbalanced labels
- Understand the different measures computable from a confusion matrix
- Be aware that each of these measures has a variety of names

		True Class y		
		+	-	
Pred.	+	TP	FP	$PPV = \frac{TP}{TP+FP}$
$\hat{y}$	-	FN	TN	$NPV = \frac{TN}{FN+TN}$
		$TPR = \frac{TP}{TP+FN}$	$TNR = \frac{TN}{FP+TN}$	$Accuracy = \frac{TP+TN}{TOTAL}$

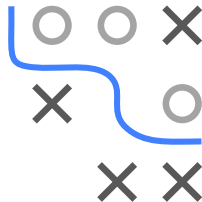
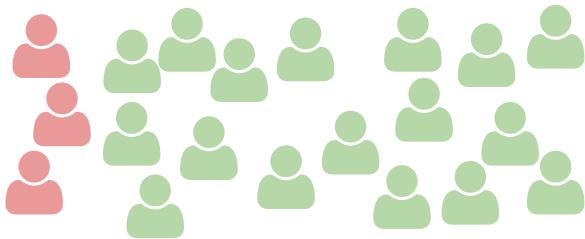
CLASS IMBALANCE I

- Assume a binary classifier diagnoses a serious medical condition.
- Label distribution is often **imbalanced**, i.e, not many people have the disease.
- Evaluating on mce is often inappropriate for scenarios with imbalanced labels:
 - Assume that only 0.5 % have the disease.
 - Always predicting “no disease” has an mce of 0.5 %, corresponding to very high accuracy.
 - This sends all sick patients home → bad system
- This problem is known as the **accuracy paradox**.



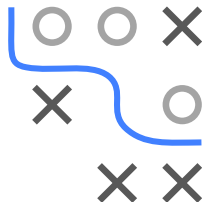
CLASS IMBALANCE II

Classifying all observations as “no disease” (green) yields top accuracy simply because the “disease” occurs so rarely → accuracy paradox.



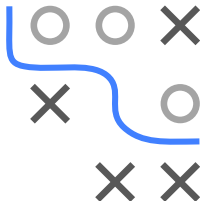
IMBALANCED COSTS I

- Another point of view is **imbalanced costs**.
- In our example, classifying a sick patient as healthy should incur a much higher cost than classifying a healthy patient as sick.
- The costs depend a lot on what happens next: we can well assume that our system is some type of screening filter, and often the next step after labeling someone as sick might be a more invasive, expensive, but also more reliable test for the disease.
- Erroneously subjecting someone to this step is undesirable (psychological, economic, medical expense), but sending someone home to get worse or die seems much more so.
- Such situations not only arise under label imbalance, but also when costs differ (even though classes might be balanced).
- We could see this as imbalanced costs of misclassification, rather than imbalanced labels; both situations are tightly connected.



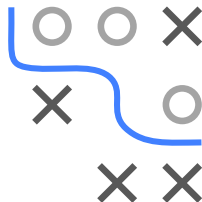
IMBALANCED COSTS II

Imbalanced costs: classifying incorrectly as “no disease” incurs very high cost.



- Problem: if we were able to specify costs precisely, we could evaluate or even optimize on them.
- This important subfield of ML is called **cost-sensitive learning**, which we will not cover in this lecture unit.
- Unfortunately, users find it notoriously hard to come up with precise cost figures in imbalanced scenarios.
- Evaluating “from different perspectives”, with multiple metrics, often helps to get a first impression of system quality.

LABELS: ROC METRICS I



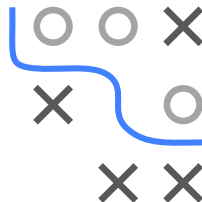
		True Class y		
		+	−	
Pred.	+	TP	FP	$\rho_{PPV} = \frac{TP}{TP+FP}$
$\hat{y}$	−	FN	TN	$\rho_{NPV} = \frac{TN}{FN+TN}$
		$\rho_{TPR} = \frac{TP}{TP+FN}$	$\rho_{TNR} = \frac{TN}{FP+TN}$	$\rho_{ACC} = \frac{TP+TN}{TOTAL}$

- True positive rate ρ_{TPR} : how many of the true 1s did we predict as 1?
- True Negative rate ρ_{TNR} : how many of the true 0s did we predict as 0?
- Positive predictive value ρ_{PPV} : if we predict 1, how likely is it a true 1?
- Negative predictive value ρ_{NPV} : if we predict 0, how likely is it a true 0?
- Accuracy ρ_{ACC} : how many instances did we predict correctly?

LABELS: ROC METRICS I

Example:

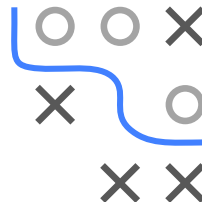
		Actual Class y		
		Positive	Negative	
$\hat{y}$ Pred.	Positive	True Positive (TP) = 20	False Positive (FP) = 180	Positive predictive value = $TP / (TP + FP)$ = $20 / (20 + 180)$ = 10%
	Negative	False Negative (FN) = 10	True Negative (TN) = 1820	Negative predictive value = $TN / (FN + TN)$ = $1820 / (10 + 1820)$ ≈ 99.5%
		True Positive Rate = $TP / (TP + FN)$ = $20 / (20 + 10)$ ≈ 67%	True Negative Rate = $TN / (FP + TN)$ = $1820 / (180 + 1820)$ = 91%	



https://en.wikipedia.org/wiki/Receiver_operating_characteristic

MORE METRICS AND ALTERNATIVE TERMINOLOGY I

Unfortunately, for many concepts in ROC, 2-3 different terms exist.



		True condition			
		Total population		Prevalence $= \frac{\sum \text{Condition positive}}{\sum \text{Total population}}$	Accuracy (ACC) = $\frac{\sum \text{True positive} + \sum \text{True negative}}{\sum \text{Total population}}$
Predicted condition	Predicted condition positive	Condition positive True positive, Power	Condition negative False positive, Type I error	Positive predictive value (PPV), Precision = $\frac{\sum \text{True positive}}{\sum \text{Predicted condition positive}}$	False discovery rate (FDR) = $\frac{\sum \text{False positive}}{\sum \text{Predicted condition positive}}$
	Predicted condition negative	False negative, Type II error	True negative	False omission rate (FOR) = $\frac{\sum \text{False negative}}{\sum \text{Predicted condition negative}}$	Negative predictive value (NPV) $= \frac{\sum \text{True negative}}{\sum \text{Predicted condition negative}}$
		True positive rate (TPR), Recall, Sensitivity, probability of detection $= \frac{\sum \text{True positive}}{\sum \text{Condition positive}}$	False positive rate (FPR), Fall-out, probability of false alarm $= \frac{\sum \text{False positive}}{\sum \text{Condition negative}}$	Positive likelihood ratio (LR+) $= \frac{\text{TPR}}{\text{FPR}}$	Diagnostic odds ratio (DOR) $= \frac{\text{LR+}}{\text{LR-}}$ F ₁ score = $\frac{1}{\frac{1}{\text{Recall}} + \frac{1}{\text{Precision}}}$
		False negative rate (FNR), Miss rate $= \frac{\sum \text{False negative}}{\sum \text{Condition positive}}$	Specificity (SPC), Selectivity, True negative rate (TNR) $= \frac{\sum \text{True negative}}{\sum \text{Condition negative}}$	Negative likelihood ratio (LR-) $= \frac{\text{FNR}}{\text{TNR}}$	

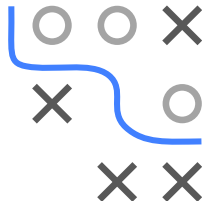
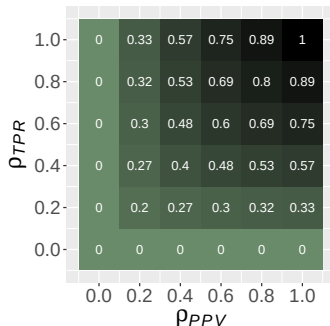
► Clickable version/picture source

► Interactive diagram

LABELS: F_1 MEASURE II

F_1 score for different combinations of ρ_{PPV} & ρ_{TPR} .

→ Tends more towards the lower of the two combined values.

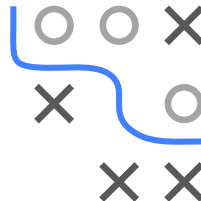


- A model with $\rho_{TPR} = 0$ (no positive instance predicted as positive) or $\rho_{PPV} = 0$ (no true positives among the predicted) has $\rho_{F_1} = 0$.
- Always predicting “negative”: $\rho_{F_1} = 0$.
- Always predicting “positive”:
$$\rho_{F_1} = 2 \cdot \rho_{PPV} / (\rho_{PPV} + 1) = 2 \cdot n_+ / (n_+ + n),$$
which will be small when the size of the positive class n_+ is small.

WHICH METRIC TO USE? I

- As we have seen, there is a plethora of methods.
→ This leaves practitioners with the question of which to use.
- Consider a small benchmark study.
 - We let k -NN, logistic regression, a classification tree, and a random forest compete on classifying the `credit_risk` data.
 - The data consist of 1000 observations of borrowers' financial situation and their creditworthiness (good/bad) as target.
 - Predicted probabilities are thresholded at 0.5 for the positive class.
 - Depending on the metric we use, learners are ranked differently according to performance (value of respective performance measure in parentheses):

metric	TPR	2 (0.8777)	3 (0.8647)	1 (0.9257)	4 (0.8357)
	TNR	4 (0.3764)	2 (0.4797)	3 (0.4072)	1 (0.4911)
	PPV	4 (0.7665)	1 (0.7947)	3 (0.7842)	2 (0.7925)
	F1	3 (0.8179)	2 (0.8279)	1 (0.8488)	4 (0.8130)
	AUC	4 (0.7092)	2 (0.7731)	1 (0.7902)	3 (0.7293)
	ACC	4 (0.7270)	2 (0.7490)	1 (0.7700)	3 (0.7320)
		k-NN	logistic regression	random forest	CART



WHICH METRIC TO USE? II

- We need not expect overly large discrepancies in general, but neither will we always see an unambiguous picture.
- Different metrics emphasize different aspects of performance.
→ The choice should be made in the domain context.
- For practitioners it is vital to understand what should be evaluated exactly, and which measure is appropriate.
 - Regarding credit risk, for instance, defaults are to be avoided, but not at all cost.
 - The bank must undertake a certain risk to remain profitable, so a more balanced measure such as the F_1 score might be in order.
 - On the other hand, a system detecting weapons at an airport should be able to achieve very high true positive rates, even if this comes at the expense of some false alarms.

